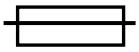
11 PLUS PREPARATION

NON-VERBAL REASONING PAPER

Name:	
Date of Birth:	
Today's Date:	

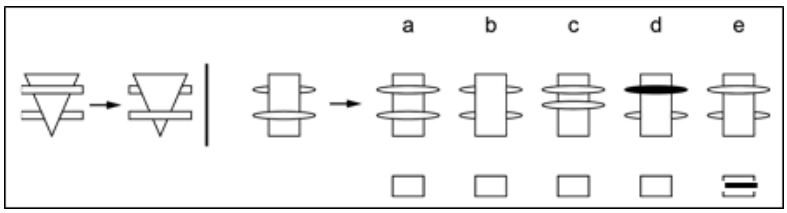
READ THESE INSTRUCTIONS CAREFULLY BEFORE STARTING.

- You have 15 minutes to complete the paper.
- Mark your answer like this: 
- This paper features 13 different question types. An example is given at the start of each question type.
- Each question is worth one mark. If you can't do a question, then move on.
- If you finish early check your answers for mistakes.
- Once the test has started you may not ask for help.

1. LIKE SHAPES / TYPE 1

Work out which of the answer shapes is related to the question shape in the same way that the example shapes are related.

Example



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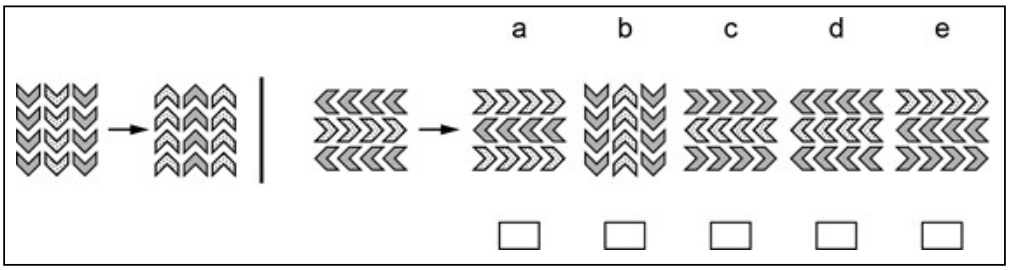

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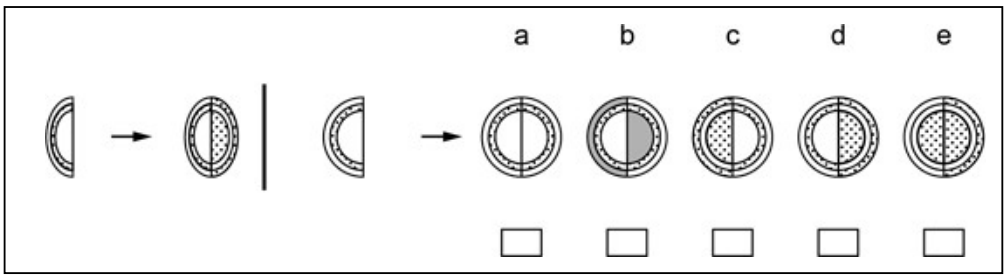

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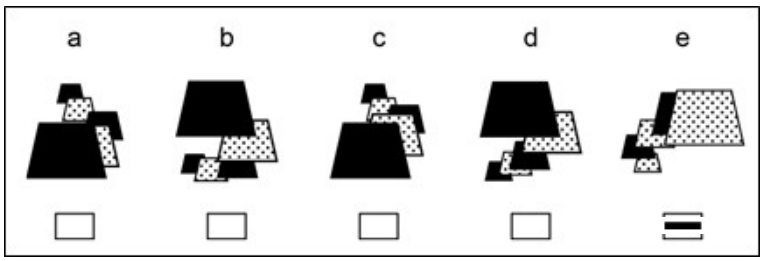
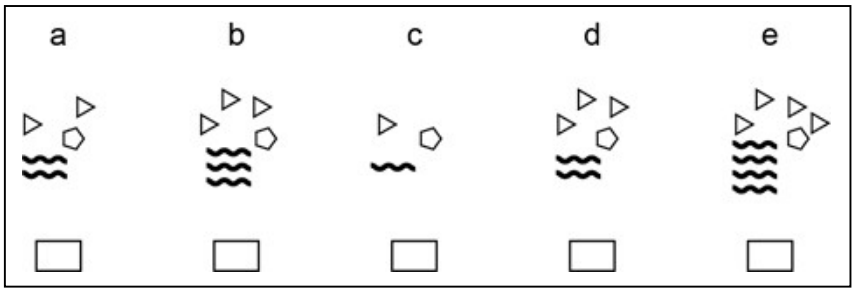
2. ODD ONE OUT / TYPE 2

Work out which of the shapes is different to the rest.

Example



1



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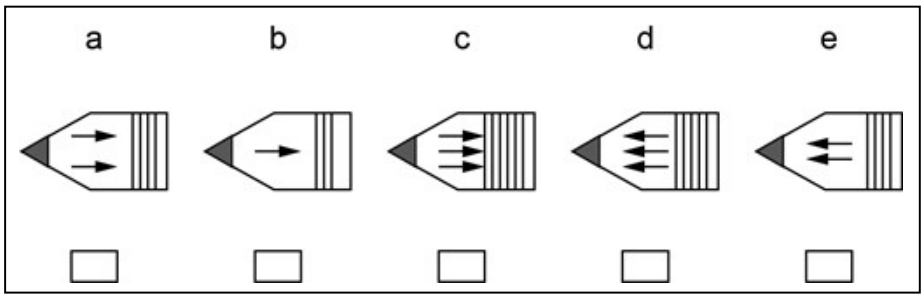
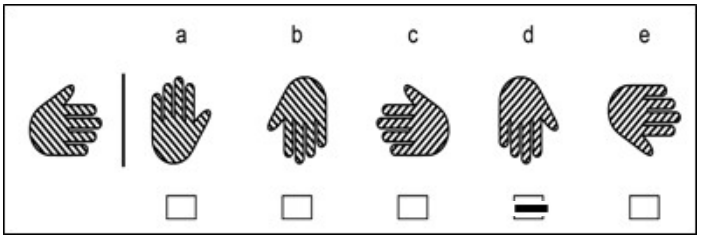


3. ROTATIONS / TYPE 3

Work out which of the answers is a rotation of the question shape.

Example



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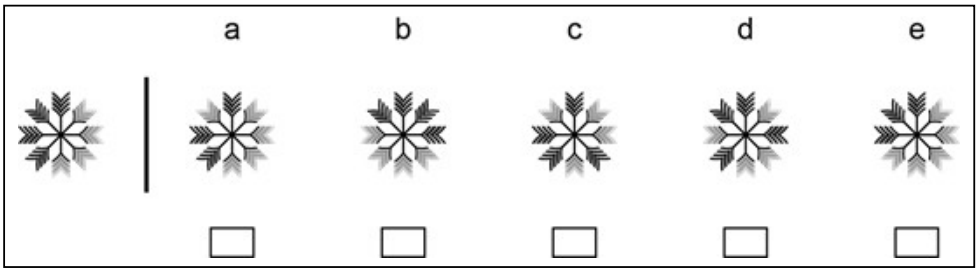

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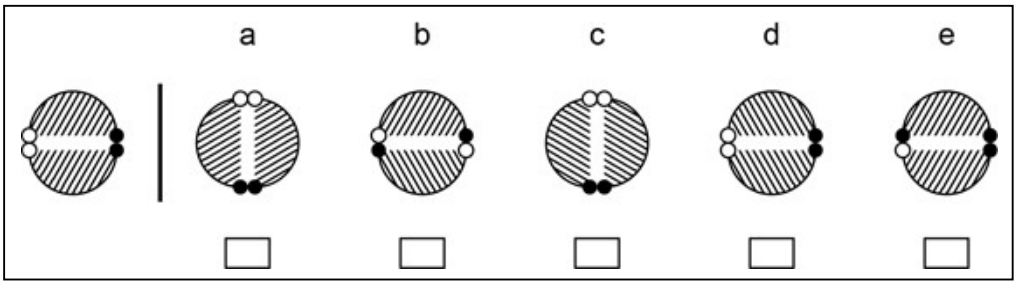
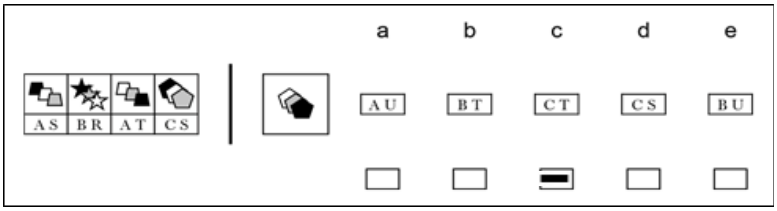


4. CODE BREAKER / TYPE 4

Look at the sequence to work out how the codes match the shapes, then select the correct answer code to match the question shape.

Example



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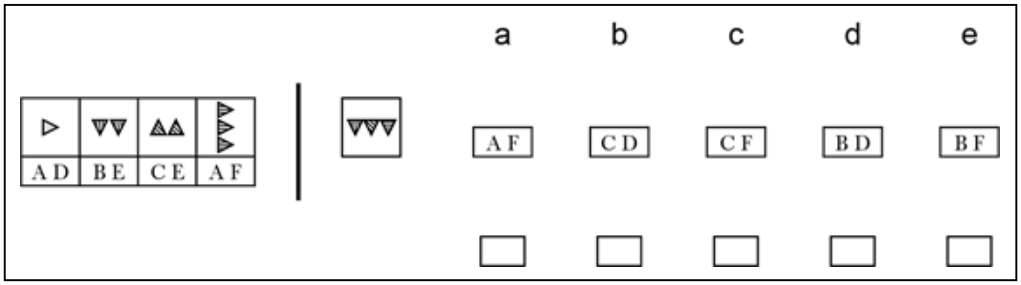

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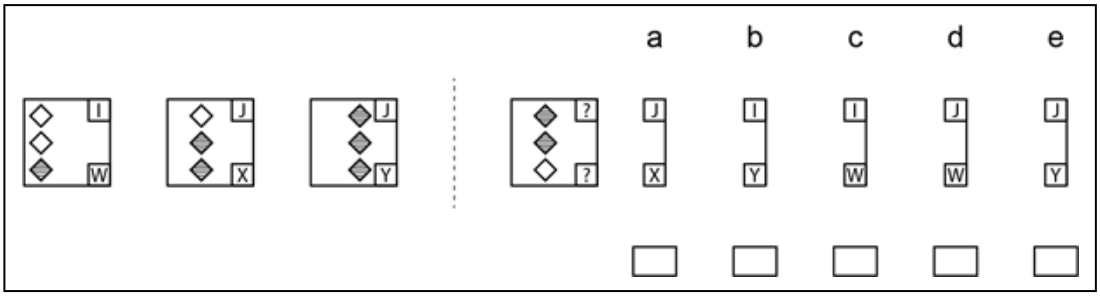
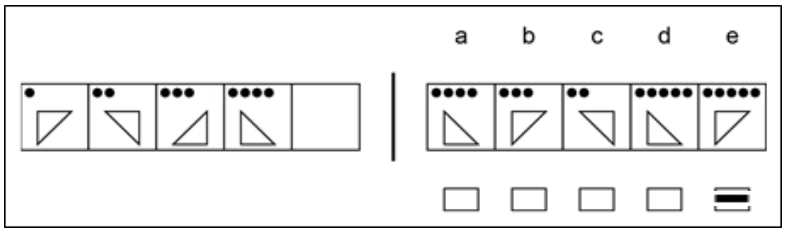


5. MISSING SEQUENCE / TYPE 5

Select the answer that correctly completes the sequence.

Example



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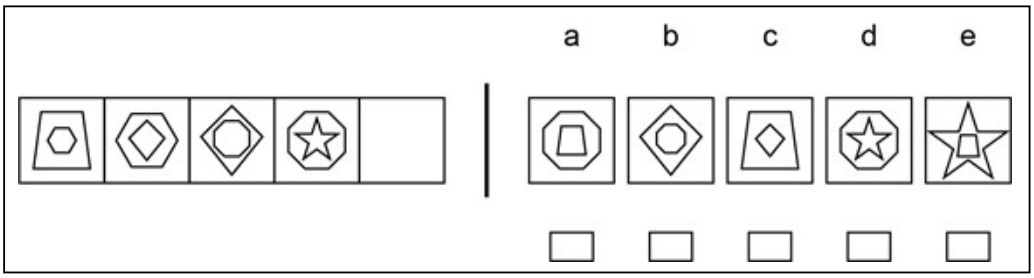

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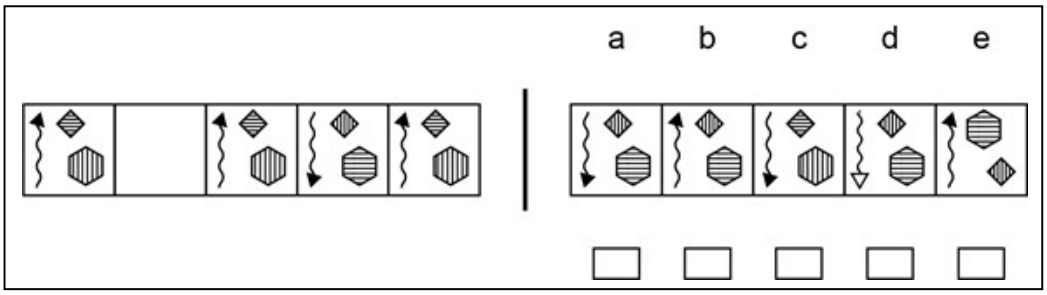

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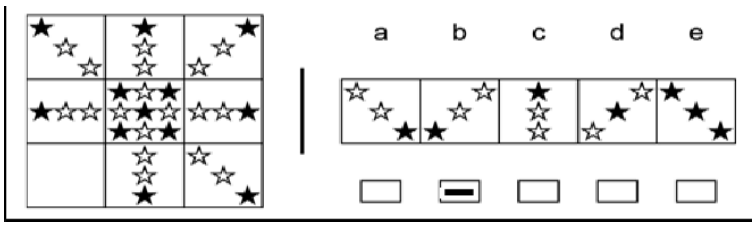
6. MISSING SQUARE / TYPE 6

Select the answer that correctly completes the grid.

Example



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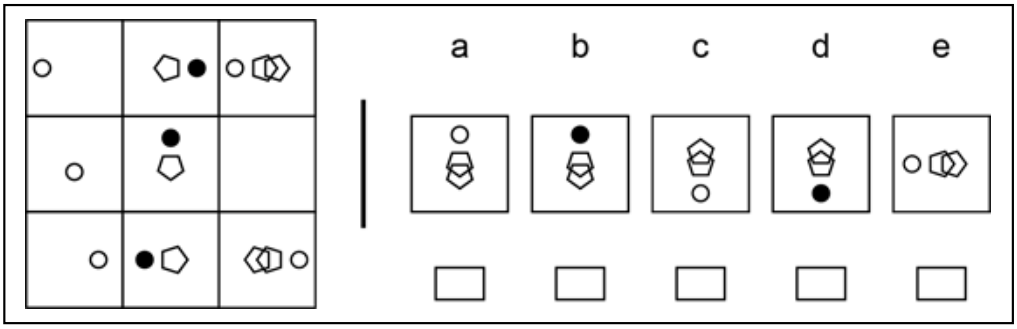

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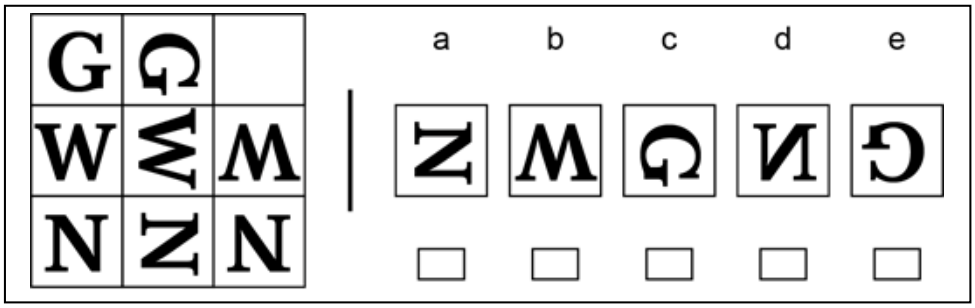
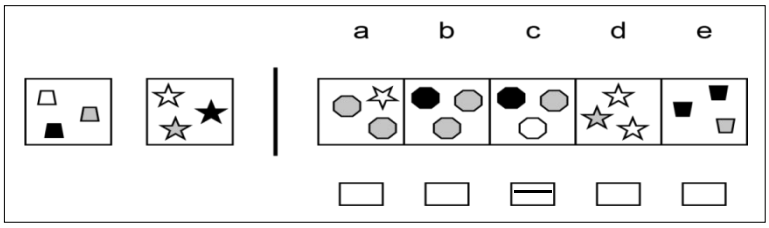


7. MOST LIKE / TYPE 7

Select the answer that is most similar to the two given shapes.

Example



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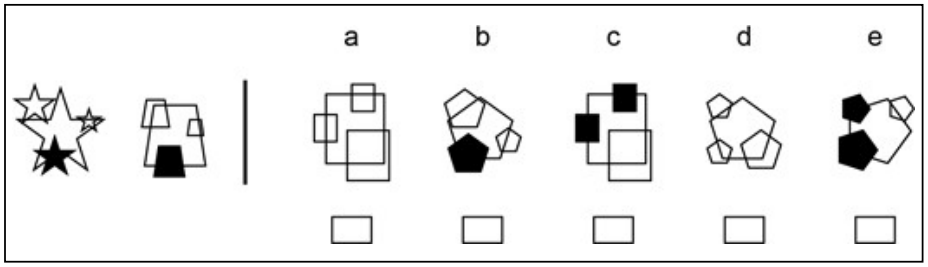

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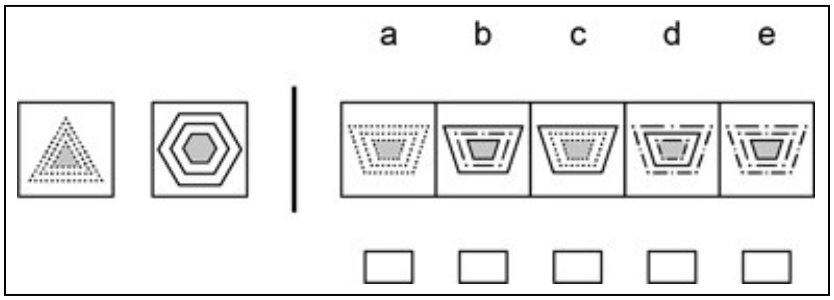

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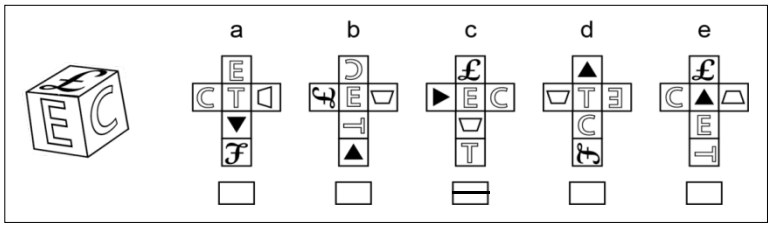
8. NETS / TYPE 8

Select the cube that is made from the given net, or net that makes the given cube.

Example



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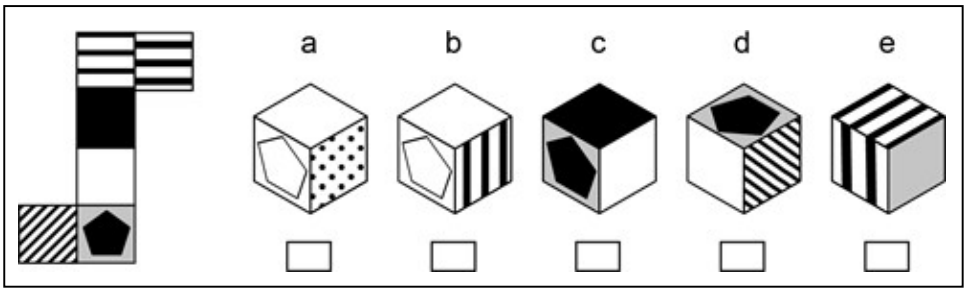

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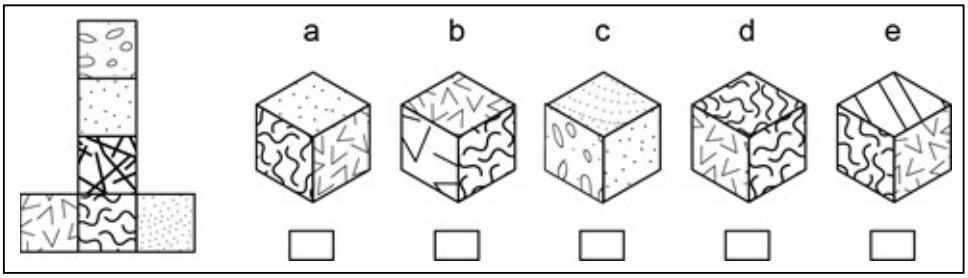
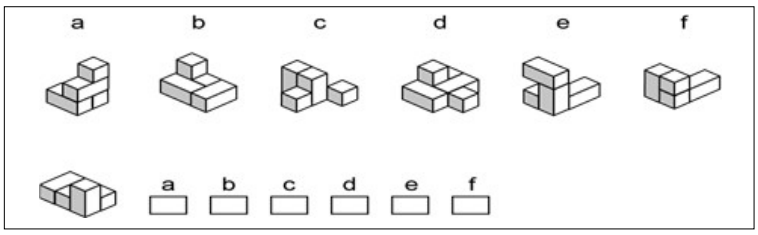


9. 3D SHAPES / TYPE 9

You will need to do one of the following:
Select the answer that is a rotation of the given 3D shape; or
Select the building blocks that make the given 3D shape; or
Select the correct 2D view of the given 3D shape.

Example



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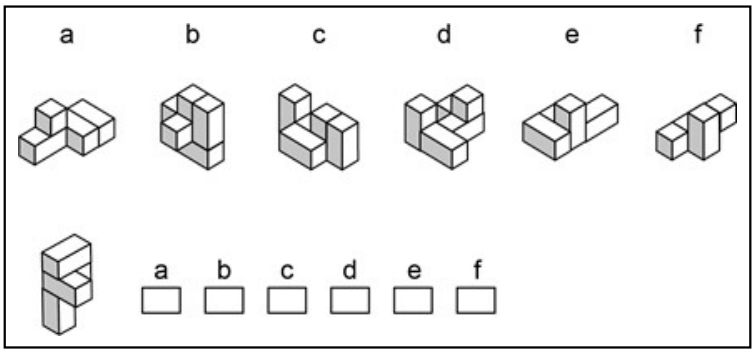
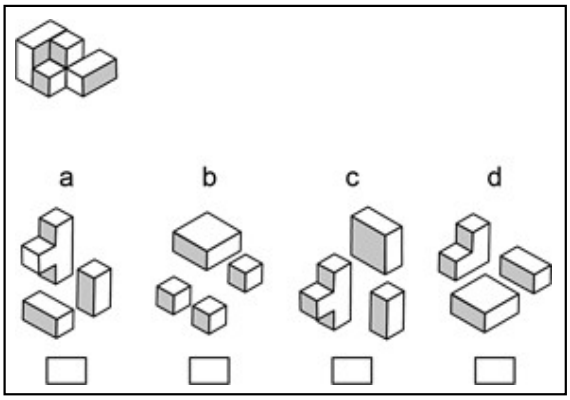


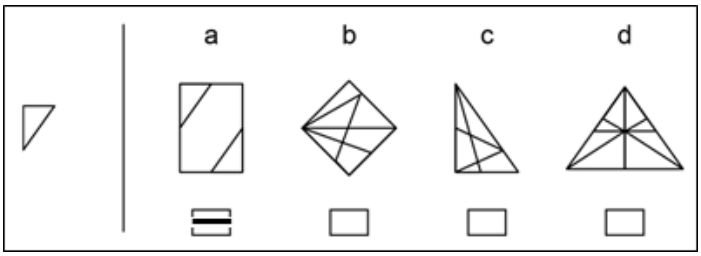
2



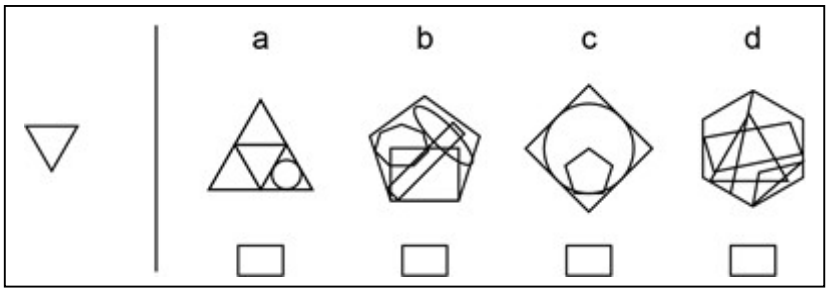
10. HIDDEN SHAPE / TYPE 10

Find the answer option that contains the shape hidden in exactly the same size and orientation.

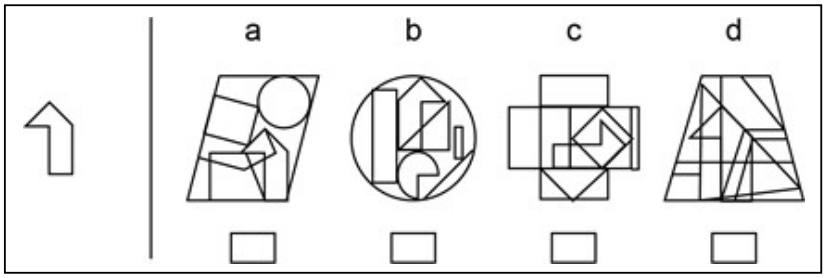
Example



1



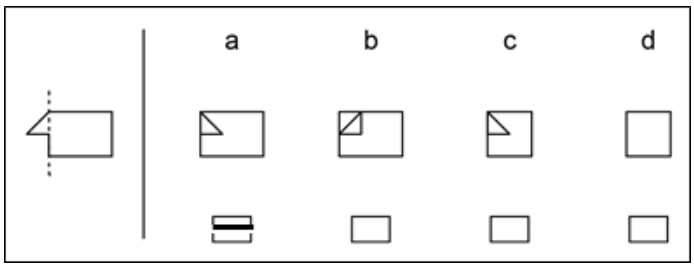
2



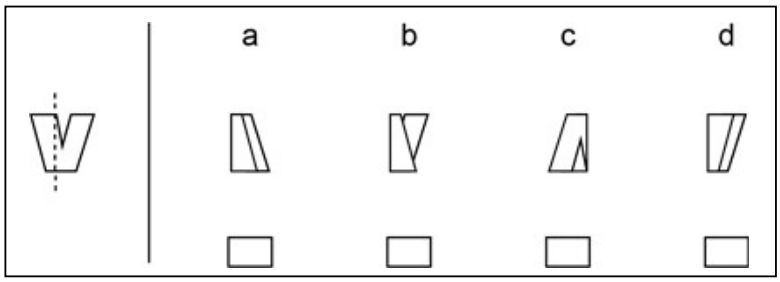
11. FOLD ALONG THE DOTTED LINE / TYPE 11

Select the option that will be formed when the shape is folded along the dotted line.

Example



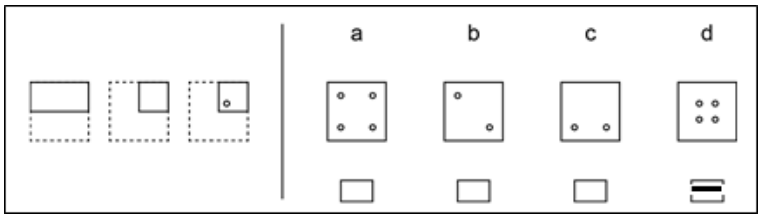
1



12. HOLE PUNCH / TYPE 12

The sequence shows a shape being folded and then hole punched. Select the option that will be created when the hole punched shape is unfolded.

Example



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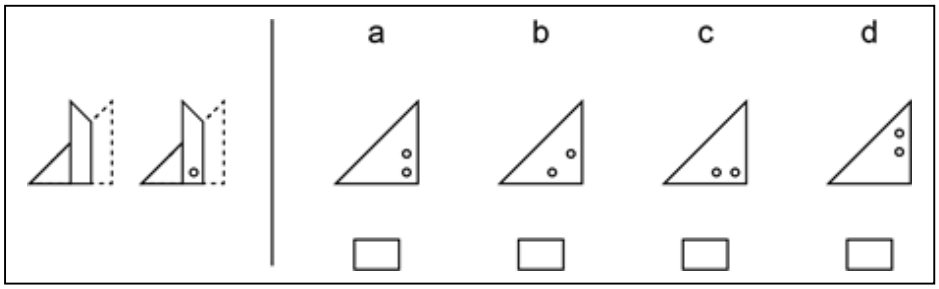
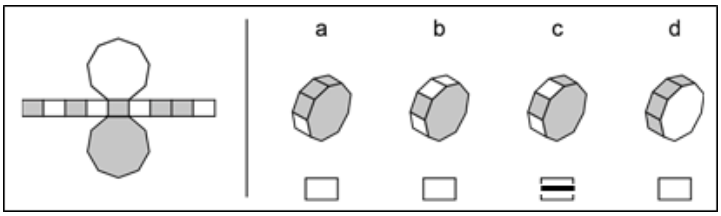


13. SHADED NETS / TYPE 13

Select the 3D shape that is created from the shaded net.

Example



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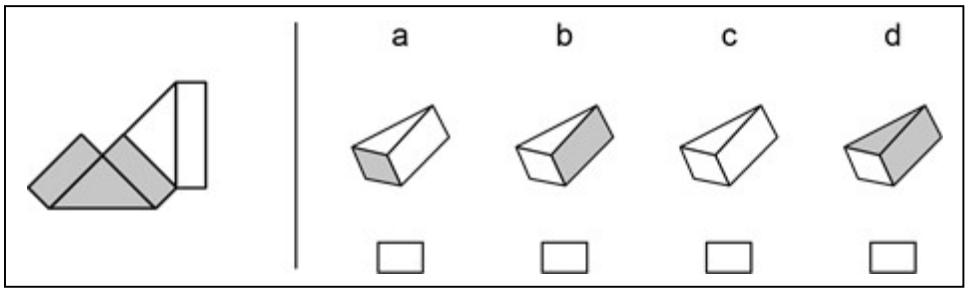


Non-Verbal Reasoning

Like Shapes

1. **A** - (The arrows are rotated 180 degrees and the dotted shading changes to light shading and the light shading changes to dotted shading.)
2. **D** - (The half shape is reflected to form a whole shape, but the colours in the new half have changed from white to dotted shading and dotted shading to white.)

Odd One Out

1. **D** - (There are the same number of wavy lines as triangles in all except D which has 2 wavy lines and 3 triangles.)
2. **C** - (There are two more vertical lines than there are arrows in all the answers except for C where there are 6 vertical lines and 3 arrows.)

Rotations

1. **B** - (B is a rotation of 90 degrees clockwise of the original.)
2. **C** - (C is a rotation of 90 degrees clockwise of the original.)

Code Breaker

1. **E** - (You can work out that A stands for the triangles pointing to the right because there are two of them. Therefore, B stands for the triangles pointing down and C stands for the triangles pointing up. In the right hand letters there are two Es and the similarity between the two is that there are two triangles in each square. So, F stands for three triangles in a square. So, the final answer is BF.)
2. **A** - (This is a very tricky question. You can work out that J stands for the middle square being shaded as there two of them and it is the only similarity. Therefore, this means that I stands for the middle square not being shaded. All of the bottom letters are different and there are two differences between all of them. Firstly, the three squares have a different number of stars shaded. So, W is one, X is two and Y is three. Or you could say W is the stars being on the left hand side, X in the middle and Y on the right. Whichever of the two that you use, the answer is J, for the middle square being shaded, and X for there being two stars shaded and being in the centre of the square. So, the final answer is JX.)

Missing Sequence

1. **E** - (As you move from left to right the inner shape becomes the outer shape in the next square with a new inner shape. As in the fourth square the inner shape is a star, in the missing square the outer shape must be a star and as the only answer with the outer shape being a star is E, this must be the answer.)
2. **A** - (The squares keep alternating with three changes each time. The arrow changes direction from up to down or down to up, the hexagon changes from vertical lines to horizontal lines or horizontal lines to vertical lines and the diamond changes from horizontal lines to vertical lines or vertical lines to horizontal lines. So, the answer must be A.)

Missing Square

1. **C** - (As you move across the two completed rows from left to right, a number of things happen: the small circle alternates between white and black, the number of pentagons increases by one and the entire square is rotated 180 degrees. So the missing square should have a white circle, have three pentagons and should be pointing upwards. The answer is therefore C.)
2. **E** - (In each row, as you move from left to right the letters are rotated 90 degrees clockwise. So the missing square should be a rotation of 90 degrees clockwise of the middle square in the top row. The answer is therefore E.)

Most Like

1. **B** - (The two sets of shapes both consist of one big shape with two smaller white copies on the edge, of different sizes, and one black copy on the edge, the same size as the larger of the two white copies. The only answer that is like this is B.)
2. **A** - (The three shapes in both the squares on the left have the same type of outlines and a lightly shaded inner shape- the left square has three shapes with a dashed outline and a lightly shaded inner shape and the right square has three shapes with a solid line and a lightly shaded inner shape. The only answer with three shapes with the same outline and a lightly shaded inner shape is A as it has three trapeziums with dashed outlines and a lightly shaded inner trapezium.)

Nets

1. **D** - (From the net you can see that the white square has a lightly shaded square with a pentagon in below it (with the top of the pentagon lining up with the border with the white square). The lightly shaded square with the pentagon in also has a square with diagonal lines in to its left, with the lines running from the bottom left to the top right. From this information, you can see that the answer is D.)
2. **C** - (From the net you can see that the square with the fewest amount of dots in has the square with the circle/ellipse type shapes in to its left. When the net is folded, both these squares will have the square with the most dots in above them. From this information, you can see that the answer is C.)

3D Shapes

1. **E** - (You can see that E has rotated 90 degrees anti-clockwise in the y-axis and then 90 degrees clockwise in the z-axis.)
2. **D** - (The shape must be made from a square shaped block, an 'L' shaped block, and a cuboid.)

Hidden Shape

1. **A** -

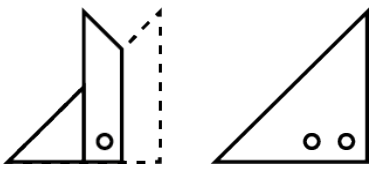
2. **A** -

Fold Along the Dotted Line

1. **D** - (The left hand portion of the shape is folded over the right hand portion so that the opposite sides do not quite line up and part of the original right hand side is still visible underneath.)

Hole Punch

1. **C** –



Shaded Nets

1. **A** - (You can see from the net that the longest rectangular face must be white, with the shorter two both shaded. A is the only option that satisfies this.)